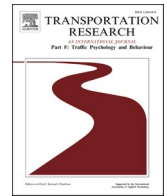




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Measuring workload effects of augmented reality head-up displays using detection response task

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ABSTRACT

Augmented reality (AR) technology could establish direct relationships between displayed information and objects in the real driving environment, e.g. by highlighting relevant objects in the traffic environment. However, it is unclear how these potential benefits of augmentation affect drivers' distraction from the driving task and their level of workload. Since gaze detection cannot separate glances at AR-elements from glances at the driving scene, new evaluation criteria are needed for measuring the distraction potential of AR-elements. The paper evaluates a methodology based on the Detection Response Task (DRT) for measuring the distraction and workload effects of three different display technologies: Head-Down Display (HDD), Head-Up Display (HUD), and Augmented-Reality HUD (AR-HUD). The DRT measures cognitive and visual workload based on a recognition-response task using the two parameters, reaction time and misses. Two experimental studies using driving simulation ($N = 24$) and production vehicles in real traffic ($N = 24$) are reported. As use cases, the subjects had to navigate through an urban traffic environment and pass different intersections. During that they were assisted by navigation systems based on the above mentioned display technologies. The resulting workload is compared to a baseline drive without the need to navigate and a reference drive in which the subjects had to work on a secondary task while driving. The results of the DRT are compared with subjective measures, gaze behaviour, and driving performance. In addition, drivers' preference and user experience regarding the three display technologies and their feedback concerning the DRT method were assessed. The results of both experimental studies show that drivers prefer the AR-HUD display technology. Reasons are mainly the hedonic aspects of the AR-HUD such as the novelty of the technology. Nevertheless, drivers report that AR-elements could obstruct important areas of the road and distract more from the driving task than HUDs. The subjective and visual-cognitive workload measured by the DRT do not differ between the HMI conditions. An analysis for each type of intersection yields most misses for the AR-HUD condition and fewest misses in the HDD condition. However, visual workload (i.e. eyes-off-road time) is higher for HDD compared to (AR-)HUD. Feedback by participants reveals that the DRT as a method for workload assessment is well accepted. All in all, the two DRT measures, misses and reaction time, allow a valid and objective workload measurement and could serve as a supplement or substitute for gaze analysis. The discussion gives indications for the measurement of workload effects caused by AR-HUD using DRT.

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1. Introduction

1.1. Advantages and disadvantages of Head-Up displays (HUD) and Augmented-Reality Head-Up displays (AR-HUD)

The design of automotive Human-Machine-Interfaces (HMIs) is a key challenge for human-factors experts, especially as vehicles are going to be equipped with more and more technologies and functions (Campbell et al., 2016; Green et al., 1994; Rozhdestvenskiy & Bouchner, 2017). With the help of HUD technology, information relevant for driving can be displayed in such a way that the driver does not have to turn away from the traffic scene. Several studies evaluated the effects of HUDs, which display the current speed, trigger warnings, and allow the interacting with a menu. The results showed that HUDs lead to reduced times needed for scanning the speedometer (Kiefer, 1991, 1998), a higher frequency of checking the current speed and fewer speeding violations when driving with HUD (Sprengr, 1993). Warnings given in the HUD result in faster reaction times (Horrey et al., 2003; Kim et al., 2013) and an improved understanding of the warning (Lorenz et al., 2014). Kiefer (1999) found fewer missed pedestrians in the HUD condition (compared to Head-Down Display, HDD) and faster reaction times for typical dangerous situations (i.e. when drivers had to recognize a pedestrian).

Miličić et al. (2008) and Miličić (2009) studied if drivers could also interact with a menu which is presented in the HUD. The results showed a faster task handling with the HUD, no differences in driving performance, and less distraction and a more efficient interaction when processing tasks in the HUD compared to a Central Information Display (CID) condition. Miličić (2009) reported that drivers show comparable compensation strategies with regard to the adaptation of the operating and driving behaviour in the HUD and CID condition.

Park and Im (2020) reported a greater effect of the number of information elements than of the position of the HUD or the arrangement of the HUD elements on workload and drivers' reaction times. Concerning design aspects of HUDs, the authors propose that the number of information elements displayed in the HUD should be less than six and the main information should be displayed in the center.

All in all, the analysed studies show mostly positive effects of a HUD compared to HDD and CID, e.g. in terms of improved eyes-on-road time (Kiefer & Gellatly, 1996), better situation awareness (especially when driving partially automated; Endsley, 1996; Wulf et al., 2014), shorter reaction times to sudden events (Horrey et al., 2003; Kim et al., 2013), lower workload (Park & Im, 2021), and higher sense of safety (Aydogdu et al., 2019). Furthermore, no negative effects on driving performance were observed.

One current approach using new technologies to make HMIs more user-friendly are HUDs including AR elements. Current production vehicles already use this display technology (e.g. Mercedes S-Class, Volkswagen ID.3). AR-HUDs provide the advantage of displaying information directly in the driver's line of sight – as traditional HUDs do. In addition to HUDs they also establish a direct relation between the information displayed and relevant objects in the real driving environment. By this, navigation instructions can be given, Advanced Driver Assistance System (ADAS) information displayed, and warnings (e.g. of pedestrians on the road) triggered (Gabbard et al., 2014). Several studies analysed the effects of navigation systems using AR-elements. The results showed that drivers prefer AR-elements, because they are helpful and reduce workload (Bark et al., 2014; Bauerfeind et al., 2019; Bauerfeind et al., 2021; Medenica et al., 2011), facilitate the detection of the correct way (Bark et al., 2014; Bauerfeind et al., 2019), and reduce navigation errors (Bauerfeind et al., 2019).

For assisted driving (SAE Level 1; SAE, 2021), Wang et al. (2020) reported that a HUD with an AR-element which dynamically follows the lead vehicle and indicates the set distance of the Adaptive Cruise Control (ACC) improves situation awareness and usability. Nevertheless, AR-elements could also improve subjective driving performance while driving without assistance (SAE Level 0), e.g. by indicating the driving path or braking point (Tönnis et al., 2007).

Studies showed that warnings based on AR-elements had the potential (compared to HDD and traditional HUD) to enhance the detection of potential hazards (Frémont et al., 2020; Rusch et al., 2013; Schall et al., 2013) and reduce reaction times (Kim et al., 2016; Rusch et al., 2013).

Despite these positive results and potential benefits, it is unclear how AR-HUDs affect drivers' distraction from the driving task (Gabbard et al., 2014). AR-elements which are designed wrongly could obstruct relevant areas of the traffic environment, distract drivers from the driving task, and increase drivers' level of workload. In addition to visual distraction by reading the AR information, cognitive distraction may occur due to the need to recognise and understand the meaning of the displayed information. AR-HUD technology also harbours the fundamental danger that the driver's attention is focused too much on the AR elements, real-world objects (e.g. obstacles on the road) are neglected (Rensink et al., 1997; O'Regan et al., 1999), and thus situation awareness is reduced (Sharfi & Shinar, 2014).

These unwanted and negative side effects could reduce the desired positive effects on traffic safety or even eliminate them. Therefore, it is crucial to measure the distractive potential of (AR-)HUD based display technologies.

1.2. Challenges for the measurement of workload and distraction when using AR-HUD

Since gaze detection as a standard method for measuring drivers' focus of attention (and by that measuring distraction from the road scene) cannot reliably separate glances at the AR-HUD from glances at the driving scene, gaze behaviour alone is not sufficient to assess effects of AR-HUDs on distraction (Kim & Gabbard, 2019). Therefore, new evaluation criteria are needed for measuring the distraction potential of AR-HUD concepts in the vehicle. This is the reason, why e.g. Miličić (2009) uses a peripheral detection task for measuring visual and cognitive distraction of HUDs.

An alternative to using gaze measurement is the Detection Response Task (DRT). This task is an accepted and efficient ISO standard method (ISO, 2016) for assessing cognitive and visual workload. The principle of the DRT method is based on a simple recognition-response task and allows to continuously measure the visual as well as cognitive workload of drivers (visual-cognitive workload). Relatively frequent artificial stimuli are presented with temporal uncertainty. The measurement of task performance is done using the two parameters reaction time and miss rate. Longer reaction times and more misses are indicative of higher visual-cognitive distraction (e.g. caused by a secondary task) and corresponding visual-cognitive workload. The ISO 17,488 (ISO, 2016) differentiates three DRT variants: head-mounted DRT, tactile DRT, and remote DRT. Whereas head-mounted means that a single LED is attached to a head bracket, the remote DRT uses a single LED mounted at a point near the central field of view of the driver. For studies in the driving simulator, the LED of the remote DRT could be replaced by a red circle embedded in the graphical visualisation of the road scene. The tactile DRT uses a vibration element attached to the shoulder of the driver. With all three DRT variants, drivers respond to the stimulus by pressing a micro switch attached to the finger.

According to ISO 17,488 (ISO, 2016), the DRT measures mainly effects on drivers' cognitive workload. But depending on the used DRT variant (remote or head-mounted DRT), the DRT approach could also be used to measure aspects of visual workload (e.g. if the driver has to look towards a display in order to fulfil a task or operate a system).

Many studies evaluated the DRT method for measuring driver distraction and proved the effectiveness of this method (Ranney et al., 2014; Stojmenova et al., 2017). Also the ISO 17,488 (ISO, 2016) refers many studies that proved that the DRT gives reliable and valuable results concerning visual and cognitive workload of drivers. The scientific findings clearly showed the sensitivity of the DRT metrics to the attentional effects of visual-cognitive workload (ISO, 2016). It was also proved that the method can be used economically and is suitable for different study setups (e.g. studies in driving simulators as well as in real traffic) (ISO, 2016).

As only a few authors used the DRT measures for investigating the attentional effects of (AR-)HUDs in road traffic (Miličić, 2009; Pampel et al., 2019; Topliss et al., 2018; Walter et al., 2019), several methodological issues are still not answered. This includes the usage of DRT variants in production vehicles in real traffic as well as in the driving simulator, the relationship between different measures of attention (subjective workload, eyes-off-road time, DRT reaction time, DRT miss rate), the sensitivity of the DRT measures, and the acceptance of this method by the subjects.

In the following, the DRT is used for an evaluation of the attentional effects of AR-HUDs while driving. Besides an assessment of the display technology, the focus is on methodological issues of the application of the DRT.

2. Research questions

AR-HUD technology allows to display information in such a way that the driver does not have to turn away from the driving scene. Furthermore, objects in the traffic environment can be highlighted. This allows innovative approaches to display navigation instructions, ADAS information, and warnings. However, it is unclear if these potential advantages of AR-HUD displays affect driver distraction from the driving task and increase workload.

Since gaze detection as a method for measuring driver attention and distraction cannot separate glances at the AR-HUD from glances at the driving scene, new evaluation criteria are needed for measuring the distraction potential of AR-HUDs in vehicles.

This paper presents and evaluates a DRT-based methodology for measuring the distraction and workload effects of different display technologies (HDD, HUD, AR-HUD) while navigating through an urban traffic environment. Two subject studies are carried out using driving simulation and production vehicles in real traffic. The study in the driving simulator used a remote DRT with a stimulus presented in the driving scene, the study in the production vehicles used a head-mounted DRT. The results gained by the DRT are compared with subjective measures, gaze behaviour, and driving performance.

The following research questions (RQ) are considered:

- I. Assessment of display technology:
 - a. Which display technology is preferred by drivers and what are the reasons for the preference?
 - b. Does display technology affect drivers' workload?
 - c. Are workload effects different for varying types of intersection (e.g. number of lanes, traffic regulation)?
 - d. Is there an effect of display technology on driving behaviour and performance?
- II. Application of DRT for measuring attentional effects
 - a. To what extent do the different workload metrics differ (subjective workload, eyes-off-road time, reaction time and miss rate of DRT)?
 - b. Are there any time-on-task and learning effects when reacting to the DRT and what does that mean for the interpretation of DRT measures?
 - c. Do subjects accept the DRT and which feedback is given concerning this method?

The results are used to derive lessons-learned and guidelines for researchers who would like to apply the DRT method for measuring workload while driving (e.g. in order to evaluate AR-HUD concepts). This includes the discussion of methodological aspects when using the DRT method in different settings, i.e. a controlled driving simulator setting or a study in real traffic.

3. Study 1 – Driving simulation

The goal of Study 1 was to evaluate whether the DRT is suitable for measuring the workload effects of different display technologies

in the very controlled setting of driving simulator studies. The study used a remote DRT with a visual stimulus presented in the driving scene and compared three HMI conditions (AR-HUD, HUD, and HDD) during navigating through an urban environment.

3.1. Method

3.1.1. Participants

A sample of 24 participants (10 female) took part in the study. The participants were recruited from the driver panel of the WIVW GmbH (Wuerzburg Institute for Traffic Sciences). All members of this panel had received a 5 h-training on vehicle control in the driving simulator before the current study took place. This training guaranteed that the participants were familiar with the simulator and prevented problems of simulator sickness.

All subjects had at least some experience with (advanced) cruise control in a privately used car. The participants had an average age of 43.5 years (sd = 12.9 years; min = 24 years, max = 68 years).

3.1.2. Simulator, eye-tracker, and DRT

The study was conducted in the static high-end driving simulator of the WIVW GmbH (Fig. 1). The driving simulation software SILAB® version 7.0 was used for the visualization of the environment as well as for the simulation of the surrounding traffic and the vehicle dynamics. An Opel Insignia Sports Tourer served as the mock-up of the driving simulator and the simulated vehicle was equipped with an automatic gearbox. The simulator had a 300° horizontal and 47° vertical field of vision, with five image channels, each one with a resolution of 1400x1050 pixels. The update frequency was 60 Hz. In addition, two LCD displays were representing the right and left outside mirrors. The interior mirror reflected a LCD display positioned in the trunk of the mock-up showing the scenery behind the vehicle.

The navigation instruction was presented via a display in the instrument cluster, a simulated HUD, or a simulated AR-HUD. The HUD and AR-HUD presented the information directly into the driving scene. A tablet in the centre console was used for presenting a secondary task. During the experimental drives, the experimenter was able to observe the driver and to communicate with the participant via intercom.

The gaze behaviour of the drivers was measured via a four-camera eye-tracking system (SmartEye 6.1; sampling rate 60 Hz, tracking accuracy of 0.5° under ideal conditions). Participants wearing glasses were asked to wear contact lenses to ensure a high quality of gaze data. The data was synchronized and recorded (60 Hz) with the software SILAB®.

The remote DRT was implemented according to ISO 17488, i.e. the stimulus was presented directly in the driving scene as a red dot slightly above the horizon (Fig. 2). It covered approx. 1° of the field of view. Subjects reacted to the stimulus using a micro switch button attached to the left index finger.

3.1.3. Experimental design and HMI conditions

Using a within-subject design, the participants experienced the three HMI conditions, AR-HUD, HUD, and HDD during navigating through an urban environment (Fig. 3).

Whereas the HUD presented the lane recommendations and the turn-by-turn instructions using 2D information (similar to the HDD), the AR-HUD gave lane recommendations using a dynamic element and highlighting the target lane. As soon as the driver's vehicle was on the target lane or when the distance to the intersection was below a certain level (approx. 50 m), chevrons occurred and guided the driver into the correct branch of the intersection. When driving with AR-HUD or HUD, the HDD was always active as well and presented the equivalent information.

3.1.4. Simulation drives, instructions, and procedure

The whole experimental procedure lasted two hours and could be divided into three phases:

First, the participant was welcomed and informed about the general course of the experiment. Various documents were presented and signed (e.g. informed consent, privacy statement). The participant entered the simulator, adjusted seat and inside mirror, and fastened the seat belt. The micro switch button for the DRT was attached to the left index finger of the participant and the tablet for the



Fig. 1. Mock-up and visual system of static high-end simulator (left, middle) and workplace of operator (right).



Fig. 2. Remote DRT stimulus (left) and micro switch button (right).



Fig. 3. HMI conditions AR-HUD (top), HUD (middle), and HDD (bottom) from driver’s point of view for a distance of 60 m (left) and 20 m (right) in front of intersection (calculated from the stop line; representation in instrument cluster is shown as well).

secondary task was shown to the participant. Since eye movement data were recorded, a driver-specific calibration procedure of the eye tracking system took place afterwards. The participant performed a short practice trip in the simulator in order to familiarize with the DRT and the handling of the micro switch button. The phase ended with a Baseline and a Reference drive (Table 1). Before the Reference drive started, the participant practiced the manual radio tuning task.

Second, the participant experienced the three experimental conditions (HDD, HUD, AR-HUD) in an order depending on the permutation plan. Therefore, the participant conducted the Navigation drive (Table 1) three times. After each drive, the participants gave feedback concerning their workload and filled out a questionnaire measuring user experience and subjective distraction from the driving task caused by the HMI.

Third and after experiencing all three HMI variants, the participant conducted the Baseline and the Reference drive a second time.

Moreover, a preference judgement concerning the HMI was asked for and the participant filled out a questionnaire measuring feedback concerning the DRT. Finally, the participant received a compensation and was discharged.

The Baseline and Reference drives were used to rank the performance of the participants between these two drives (NHTSA, 2013).

For some analyses, the intersections of the Navigation drive were classified according to their layout, i.e. the number of lanes and the direction of turning. The single-lane intersections were right-before-left crossings, whereas the other intersections were controlled by a traffic light.

3.1.5. Dependent measures

A number of variables from the areas (1) preference and user experience, (2) workload, (3) driving behaviour and performance, and (4) acceptance of DRT was measured (Table 2):

3.1.6. Analyses and statistics

The data were analysed both descriptively and inferential statistically. To assess the effects of the within factor ‘Drive’ (Baseline vs Reference vs HDD Navigation vs HUD Navigation vs AR-HUD Navigation) and ‘Intersection layout’ (single lane left turn vs single lane right turn vs two lanes left turn vs two lanes right turn vs three lanes left turn vs three lanes right turn), an analysis of variance (ANOVA) with repeated measurements was conducted. Scheffé tests were used for post-hoc comparisons (as this is a conservative method of post-hoc testing; Winer, 1962), if a main effect reaches statistical significance. Mean values (m) and confidence intervals (CI) were calculated for post-hoc comparisons.

Mauchly’s test was used for proving sphericity. Normal distribution of the dependent variables was assumed, since ANOVA is robust to a violation of the normal distribution assumption (Pagano, 2010). Greenhouse-Geisser adjustment was used to correct for violations of sphericity.

The measures for the Baseline drive and the Reference drive were averaged over the two repeated drives at the beginning and at the end of the experimental procedure. An alpha level of 0.05 was used to determine statistical significance.

3.2. Results

3.2.1. Preference and user experience

Participants clearly favoured the AR-HUD. After experiencing all three HMI variants, 58 % of participants preferred to be supported by an AR-HUD based navigation system. 33 % reported a preference for the HUD and 8 % for the HDD. (Fig. 4 left).

There was a significant main effect of HMI on user experience ($F(2, 46) = 10.99$, $p < .001$) with a significantly poorer user experience of the HDD compared to HUD and AR-HUD. This effect was solely due to hedonic quality ($F(2, 46) = 25.74$, $p < .001$; Fig. 4

Table 1

Description of and instruction for the experimental drives (please note: there was no vehicle directly in front of the ego-vehicle that would possibly interfere with the DRT).

Description	Instruction
Baseline drive Urban route with oncoming traffic and pedestrians on the pavement. Duration: approx. 2 min. HMI: HDD. Subjects drove the course manually and without navigation system and carried out the DRT.	“Adhere to the road traffic regulations and always drive straight ahead. Please work on the reaction task. Nevertheless, the driving task should always have the highest priority. Work on the reaction task only if this does not jeopardise your driving safety.”
Reference drive Urban route with oncoming traffic and pedestrians on the pavement. Duration: approx. 2 min. HMI: HDD. Subjects drove the course manually and without navigation system, worked on a secondary task (Manual Radio Tuning Task; Krause et al., 2015), and carried out the DRT.	“In addition to the driving and reaction task, you are now to complete another task, the so-called manual radio tuning task. This is to simulate the operation of an “old radio” where you still had to set frequencies yourself. [Detailed description of manual radio tuning task.] Adhere to the road traffic regulations and always drive straight ahead. Please work on the reaction task. Nevertheless, the driving task should always have the highest priority, followed by the manual radio tuning task. Work on the reaction task only if this does not jeopardise your driving safety.”
Navigation drive Urban route with several intersections, oncoming traffic and pedestrians on the pavement. Duration: approx. 7 min. HMI: according to experimental condition. Subjects drove the course manually and with navigation system, and carried out the DRT. The types of intersections included (a) single lane, right-before-left, (b) two lanes, traffic light controlled, (c) three lanes, traffic light controlled. As participants drove the navigation drive several times, three routes were prepared with a modified sequence of the crossing scenarios in each case.	“You now drive with the navigation system active. This means that you will receive navigation instructions in the HDD/HUD/AR-HUD [depending on HMI condition]. Follow these instructions. The navigation instruction is presented only visually, i.e. without auditory output. Please use this information and drive according to these instructions. If the navigation system does not show anything, please drive straight ahead. Please work on the reaction task. Nevertheless, the driving task should always have the highest priority. Work on the reaction task only if this does not jeopardise your driving safety.”

Table 2
Dependent variables.

Preference and user experience	Preference judgement of the HMI variants: At the end of the experiment, the participants answered the question “Which of the display variants did you prefer to drive with?” [rank]. Standardized questionnaire: The participants filled out a standardized questionnaire on user experience after each condition (UEQ-S; Schrepp, Hinderks & Thomaschewski, 2017). The questionnaire includes two subscales: pragmatic (task related aspects, e.g. efficiency, perspicuity) and hedonic quality (non-task related aspects, e.g. stimulation, novelty). Obstruction and distraction by display: After each condition, the respondents rejected or agreed on two statements concerning obstruction and distraction by the display (“The display obstructed important areas of the road.”, “The display distracted me from my driving task.”) [agreement to statement on five-point rating scale].
Workload	Standardized questionnaire: The participants filled out a standardized questionnaire measuring workload after each drive (DALI; Pauzie, 2008). Detection response task: General degree of visual-cognitive distraction measured by the parameters miss rate [% visual targets not detected] and reaction time for detected visual targets [ms]. Eyes-off-road time: Drivers’ head and eye movements were recorded with high accuracy using cameras. The portion of time the driver was not visually attentive to the road scene ahead was measured by analysing gaze behaviour [% of gaze duration away from the road].
Driving behaviour and performance	Number of false turning manoeuvres: During the Navigation drive, the experimenter observed and coded after each intersection, if the participant turned correctly [no / yes]. Distance at start of blinking: It was recorded in which distance from the intersection the participants operated the indicator [m].
Acceptance of DRT	Feedback concerning DRT: At the end of the experimental procedure, participants answered several questions regarding their experience with performing the DRT (“In general, the stimulus was badly perceptible.”, “Working on the reaction task was unpleasant.”, “I would not participate again in a similar experiment.”) [agreement to statement on five-point rating scale].

right), while pragmatic quality was not rated differently for the three HMI variants ($F(2, 46) = 2.29, p < .113$; Fig. 4 middle). Concerning hedonic quality, statistical posts-hoc testing showed that HUD was rated better than HDD and AR-HUD better than HDD and HUD ($m_{HDD} = 1.78, 95\% \text{-CI}_{HDD}[0.61, 1.74]$; $m_{HUD} = 1.80, 95\% \text{-CI}_{HUD}[1.39, 2.21]$; $m_{AR-HUD} = 1.30, 95\% \text{-CI}_{AR-HUD}[0.68, 1.92]$).

There was a significant effect of display technology on drivers’ feedback concerning obstruction ($F(2, 46) = 20.55, p < .001$; Fig. 5 left). With AR-HUD, a considerable number of drivers ($n = 7$, i.e. 29 %) rather agreed with the statement that important areas of the road are obstructed due to the displayed navigation information.

Drivers did not report a different level of distraction due to display technology ($F(2, 46) = 1.49, p = .237$; Fig. 5 right). 29 % of participants ($n = 7$) in the HDD condition, 8 % of participants in the HUD condition ($n = 2$), and 21 % of participants ($n = 5$) in the AR-HUD condition (rather) agreed with the statement that the display distracted them from the driving task.

3.2.2. Workload

The results showed a significant main effect of simulator drive on subjective workload (Greenhouse-Geisser adjustment was used to correct for violations of sphericity: $F(2.94, 67.61) = 31.87, p < .001$; Fig. 6 top left). The Reference drive was clearly more demanding than the Baseline drive. Post-hoc tests show that the workload during the Navigation drive (independent of the HMI condition) was higher than for the Baseline drive and lower than for the Reference drive. The HMI conditions (HDD, HUD, AR-HUD) did not differ in their level of subjective workload ($m_{Baseline} = 21.50, 95\% \text{-CI}_{Baseline}[15.71, 27.29]$; $m_{Reference} = 69.00, 95\% \text{-CI}_{Reference}[59.90, 78.10]$; $m_{HDD} = 46.83, 95\% \text{-CI}_{HDD}[38.61, 55.06]$; $m_{HUD} = 43.83, 95\% \text{-CI}_{HUD}[35.49, 52.17]$; $m_{AR-HUD} = 44.67, 95\% \text{-CI}_{AR-HUD}[36.92, 52.41]$).

The miss rate in the DRT also revealed a significant main effect of driving condition (Greenhouse-Geisser adjusted: $F(2.07, 47.51) = 26.50, p < .001$; Fig. 6 bottom left). There were significantly more misses during the Reference drive than during the Baseline drive (post-hoc test). The miss rate during the Navigation drive was between that of the Baseline drive and the Reference drive, i.e. the Navigation drive leads to significantly more misses than the Baseline drive but to less misses than the Reference drive ($m_{Baseline} = 0.004, 95\% \text{-CI}_{Baseline}[0.000, 0.009]$; $m_{Reference} = 0.245, 95\% \text{-CI}_{Reference}[0.177, 0.313]$; $m_{HDD} = 0.116, 95\% \text{-CI}_{HDD}[0.090, 0.143]$; $m_{HUD} = 0.138, 95\% \text{-CI}_{HUD}[0.102, 0.174]$; $m_{AR-HUD} = 0.147, 95\% \text{-CI}_{AR-HUD}[0.110, 0.184]$). Again, there was no difference between

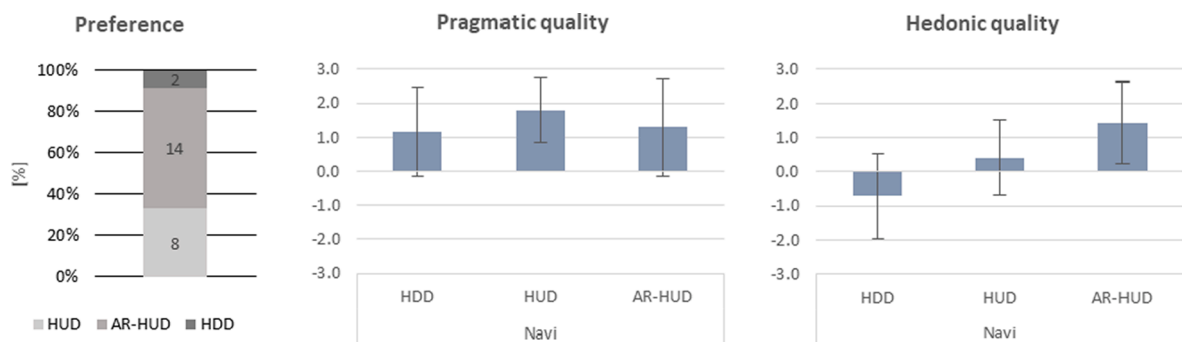


Fig. 4. Preference of HMI conditions (left), rating of pragmatic quality of user experience (middle) and hedonic quality of user experience (right) for HMI conditions (mean value $\pm$ standard deviation).

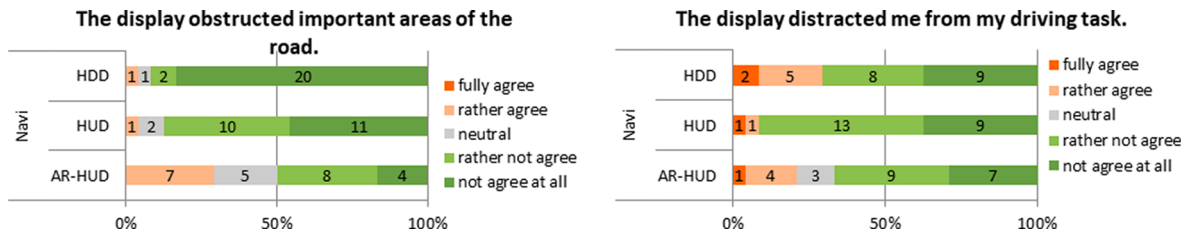


Fig. 5. Feedback of drivers concerning obstruction (left) and distraction (right) by display.

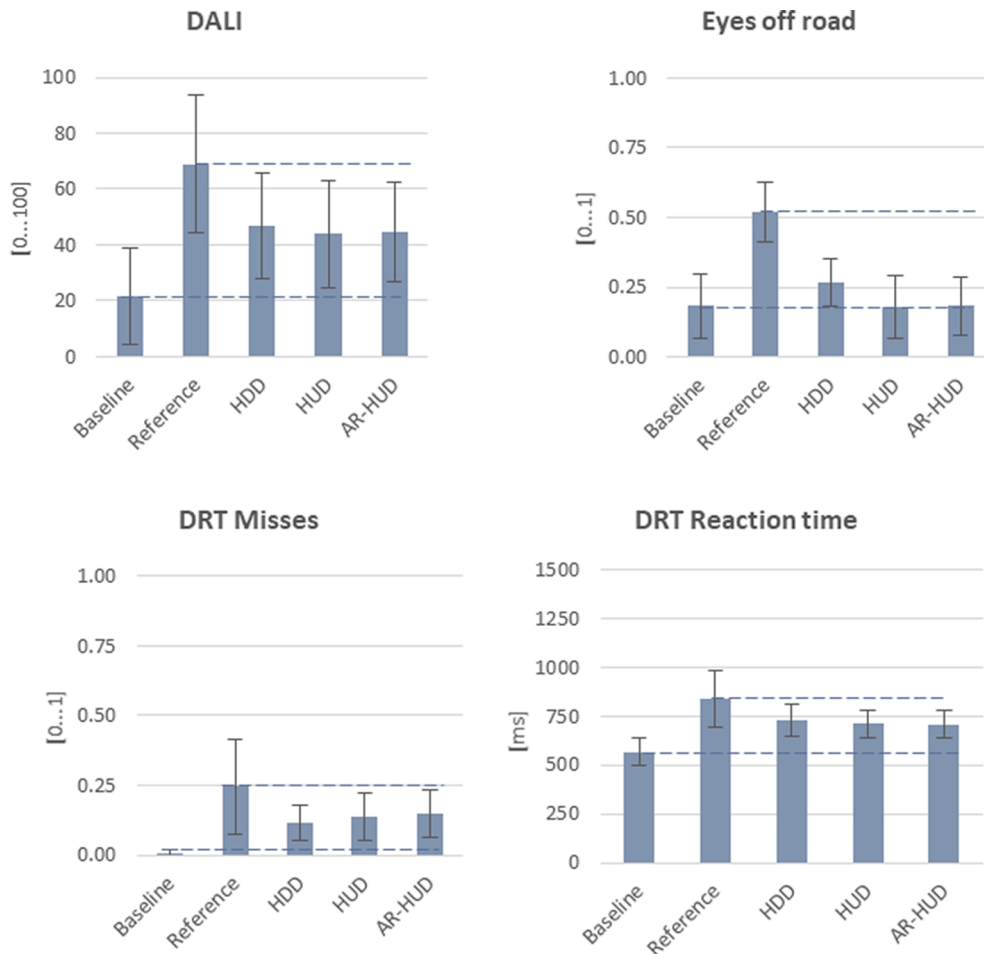


Fig. 6. Subjective workload (top left), portion of time with eyes off road (top right), DRT miss rate (bottom left), and DRT reaction time (bottom right) for Baseline drive, Reference drive, and HMI conditions in Navigation drive (mean value $\pm$ standard deviation).

the HMI conditions.

A very similar effect was seen for the reaction times in the DRT (Greenhouse-Geisser adjusted: $F(1.96, 45.14) = 64.95$, $p < .001$; Fig. 6 bottom right). Post-hoc testing revealed that the reaction times were longer for the Reference drive than for the Baseline drive with the times for the Navigation drives in between. There was no significant difference in reaction time between the HMI conditions ($m_{\text{Baseline}} = 567$, 95 %-CI_{Baseline}[542, 592]; $m_{\text{Reference}} = 840$, 95 %-CI_{Reference}[783, 897]; $m_{\text{HDD}} = 728$, 95 %-CI_{HDD}[692, 764]; $m_{\text{HUD}} = 709$, 95 %-CI_{HUD}[678, 740]; $m_{\text{AR-HUD}} = 707$, 95 %-CI_{AR-HUD}[676, 738]).

The simulator drive had also an effect on eyes-off-road time (Greenhouse-Geisser adjusted: $F(3.02, 66.34) = 87.18$, $p < .001$; Fig. 6 top right). Post-hoc testing showed that the gaze was averted from the road significantly longer during the Reference drive compared to the Baseline drive and the Navigation drives. The Navigation drive with HDD resulted in more eyes-off-road time than navigating with HUD or AR-HUD. The Navigation drives with AR-HUD and HUD did not differ with regard to eyes-off-road time ($m_{\text{Baseline}} = 0.18$, 95 %-CI_{Baseline}[0.14, 0.23]; $m_{\text{Reference}} = 0.52$, 95 %-CI_{Reference}[0.48, 0.57]; $m_{\text{HDD}} = 0.27$, 95 %-CI_{HDD}[0.23, 0.30]; $m_{\text{HUD}} = 0.18$, 95 %-CI_{HUD}[0.14, 0.23]; $m_{\text{AR-HUD}} = 0.18$, 95 %-CI_{AR-HUD}[0.14, 0.23]).

%-CI_{HUD}[0.13, 0.23]; $m_{AR-HUD} = 0.18$, 95 %-CI_{AR-HUD}[0.14, 0.23]).

In summary, the four workload measures (DALI, eyes-off-road time, DRT miss rate, DRT reaction time) showed consistently that the Navigation drive was less demanding than the Reference drive. With the exception of eyes-off-road time in HUD- and AR-HUD condition, the Navigation drive was more demanding than the Baseline drive. The subjective and visual-cognitive workload measured by the DRT did not differ between the HMI conditions. Only the eyes-off-road time yielded a higher visual workload for navigating using a HDD compared to (AR-)HUD.

The next analyses focussed on the different layouts of the intersections. The results showed that navigating with AR-HUD leads to the most DRT misses compared to fewer misses in the HUD condition and fewest errors in the HDD condition ($F(2,44) = 6.12$, $p = .005$; Fig. 7 top left). The HMI $\times$ Layout interaction did not yield statistical significance ($F(10,220) = 0.71$, $p = .716$). There was no significant effect of HMI or layout of intersection with regard to the reaction time to a DRT stimulus (Fig. 7 top right).

An analysis of gaze behaviour revealed that drivers did look more often away from the road ahead when driving in HDD condition compared to HUD and AR-HUD (Greenhouse-Geisser adjusted: $F(1.84, 33.04) = 36.59$, $p < .001$; Fig. 7 bottom). There was also a statistical significant main effect of intersection layout (Greenhouse-Geisser adjusted: $F(2.08, 37.37) = 24.76$, $p < .001$). The interaction of HMI $\times$ Layout of intersection did not yield statistical significance (Greenhouse-Geisser adjusted: $F(5.35, 96.23) = 2.11$, $p = .067$).

3.2.3. Driving behaviour and performance

When navigating in a city environment with AR-HUD, turn signals were given significantly later than with the other two HMI conditions (Greenhouse-Geisser adjusted: $F(1.41, 19.72) = 13.13$, $p < .001$; Fig. 8 left). Drivers either waited for the chevrons to occur (see Fig. 3), realised later in which direction they needed to turn, or were distracted by the AR-HUD. The effect of an earlier blinking in more complex intersections (Greenhouse-Geisser adjusted: $F(3.69, 51.67) = 143.68$, $p < 0.001$) was due to the fact that the navigation instructions were given already in a further distance in two lane intersections and – even more – in three lane intersections.

Whereas the HDD condition led to the most turning errors ($n = 10$ for all subjects; Fig. 8 right), the numbers of turning errors were lower for HUD driving ($n = 5$) and very low for the AR-HUD condition ($n = 2$). A turning error could be to turn in the wrong direction or no turning at all (and therefore driving straight ahead). As (AR-)HUDs show the navigation information in the direct field of view of the driver, it is less likely that the driver makes a wrong turning decision or misses the turning at all.

3.2.4. Acceptance of DRT

At the end of the experiment, subjects gave feedback concerning the operation of the DRT while driving (Fig. 9). Although several participants ($n = 6$; 25 %) experienced the DRT as unpleasant, all participants found the stimulus to be easily perceivable. As a

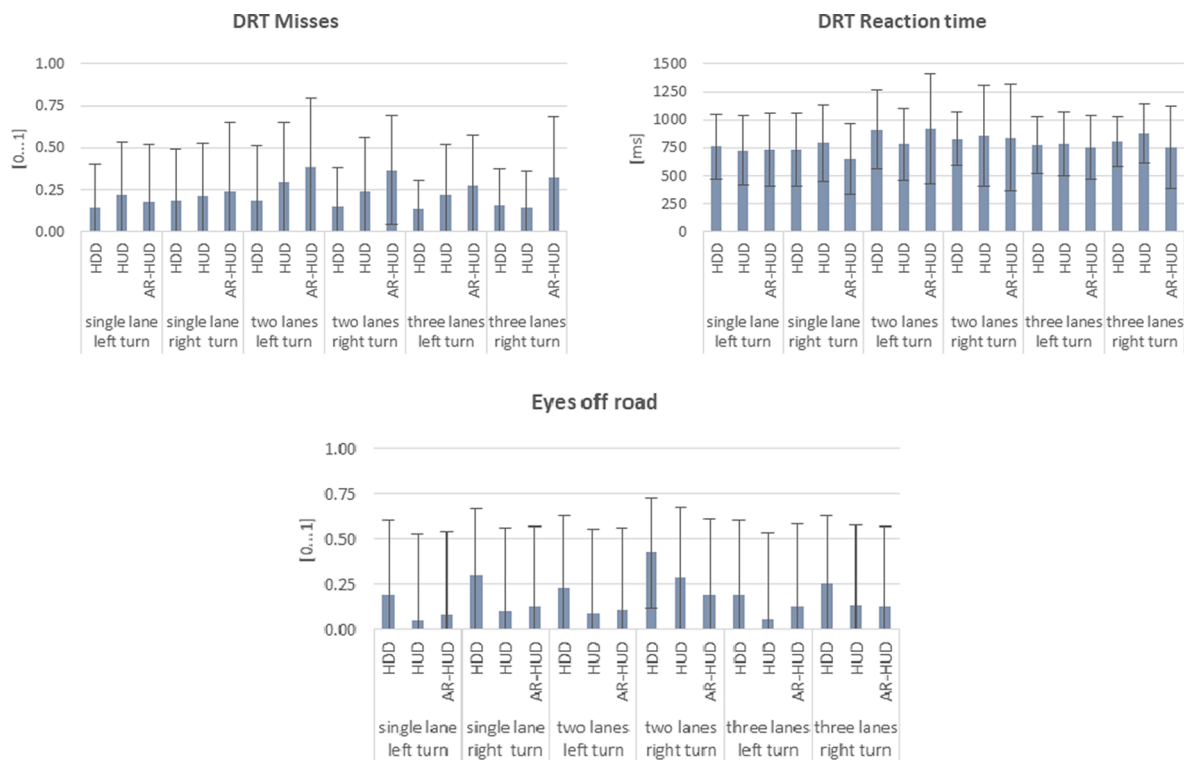


Fig. 7. DRT miss rate (top left), DRT reaction time (top right), and portion of eyes off road (bottom) for different types of intersection and HMI conditions in Navigation drive (mean value $\pm$ standard deviation).

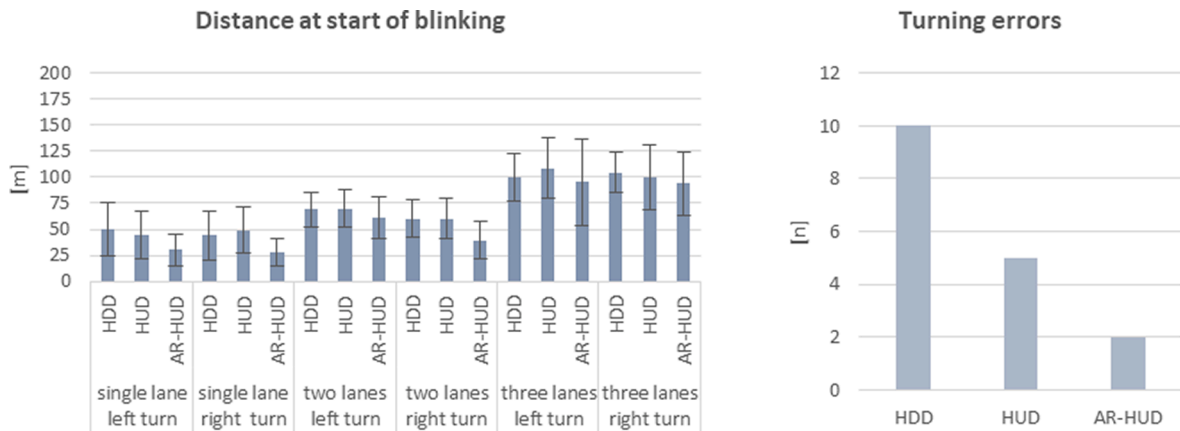


Fig. 8. Distance at start of blinking for different types of intersections and HMI conditions (mean value $\pm$ standard deviation) and number of turning errors (wrong direction, no turn at all) for HMI conditions in Navigation drive.

consequence, only one participant (4 %) would not take part in a similar test again.

4. Study 2 – Real vehicles in real traffic

The goal of Study 2 was to investigate whether the DRT can also be applied in less controlled conditions in real traffic for assessing display solutions implemented in production vehicles. The study used a head-mounted DRT with a visual stimulus and investigated two HMI conditions (AR-HUD and HUD) in two different vehicles during a navigation task in an urban environment.

4.1. Method

4.1.1. Participants

A sample of 24 participants (8 female) took part in the study. The participants were recruited from the driver panel of the WIVW GmbH (Wuerzburg Institute for Traffic Sciences). All subjects had at least some experience with (advanced) cruise control in a privately used car. The participants had an average age of 40.7 years ($sd = 15.3$ years; $min = 21$ years, $max = 70$ years).

4.1.2. Test vehicles and DRT

The study was conducted using two production vehicles: Mercedes-Benz S-Class (W223; Fig. 10 left) and Volkswagen ID.3 (Fig. 10 middle). Half of the subjects drove with the Mercedes, half with the VW. Both vehicles were equipped with an automatic gearbox. A tablet in the centre console was used for presenting a secondary task. During the experimental drives, the experimenter sat on the back seat, operated the data logging system, and instructed the driver.

The head-mounted DRT was implemented according to ISO 17488, i.e. the stimulus was a red LED attached to the subject's head by means of a holder and placed approx. 12 cm in front of the left eye (Fig. 10 right). Subjects reacted to the stimulus using a micro switch button attached to the left index finger.

4.1.3. Experimental design and HMI conditions

Using a mixed between-within-design, the participants experienced the two HMI conditions AR-HUD and HUD (within factor) in

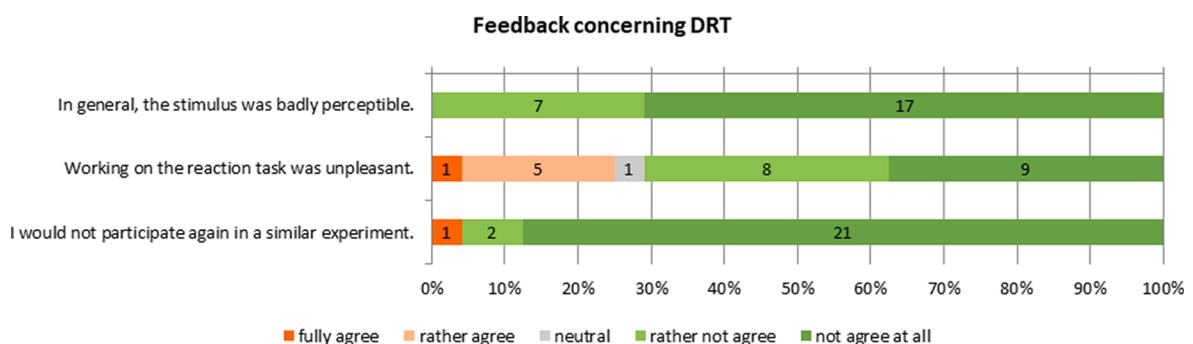


Fig. 9. Feedback of drivers concerning DRT method.



Fig. 10. Head-mounted DRT in Mercedes-Benz S-Class (left) and VW ID3 (middle), detailed view (right).

both vehicles (between factor) during a navigation drive in an urban environment.

The navigation instructions were presented either using the AR-HUD (Fig. 11) or the HUD, depending on the experimental condition. The AR navigation element in the HUD of the Mercedes S-Class (Fig. 11 left) consisted of a dynamically moving arrow. With approaching the intersection, the arrow moved towards the intersection and changed its appearance to chevrons that guided the correct turning direction. The AR navigation element in the Volkswagen ID.3 (Fig. 11 right) was more static and consisted of chevrons that became bigger when approaching the intersection and showed the turning direction.

The HUD-condition of the Mercedes S-Class presented a schematic map presentation instead of the AR element, the HUD of the VW ID.3 was identical to the AR-HUD condition without the AR element.

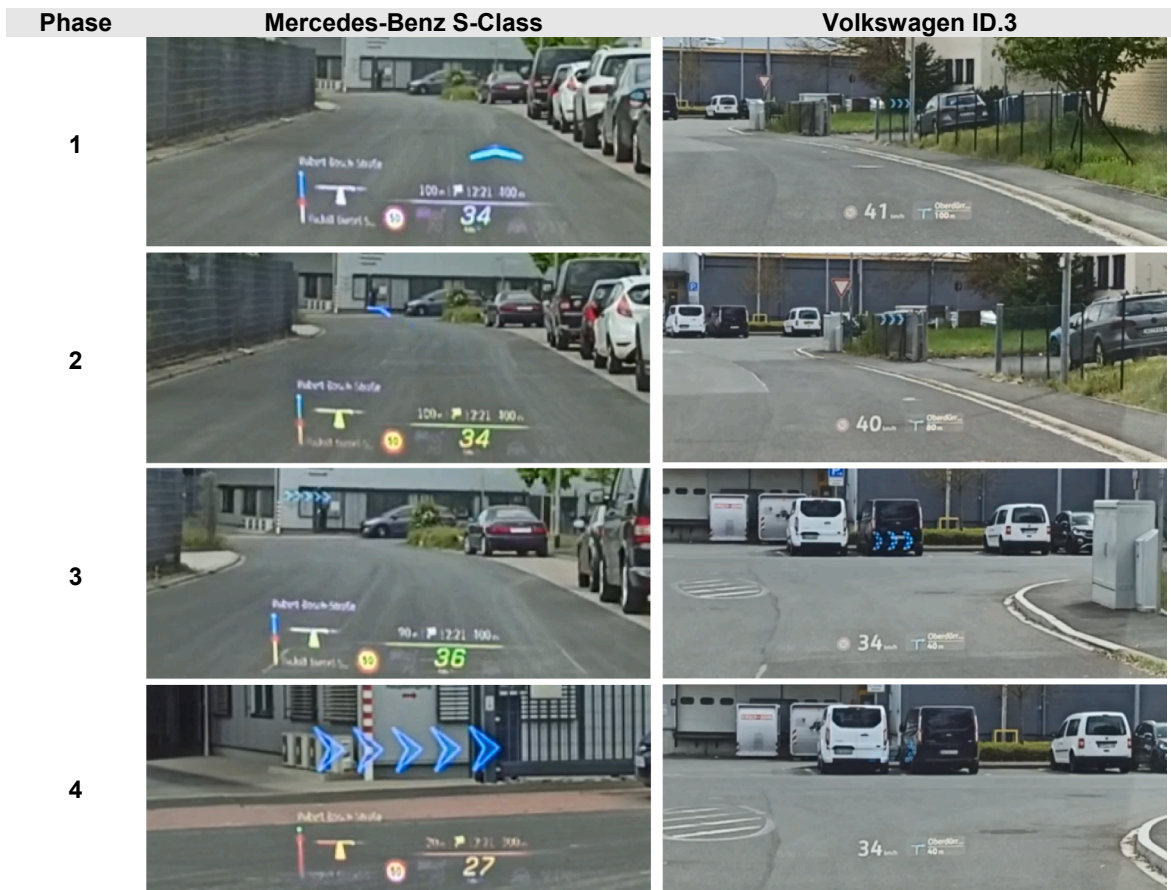


Fig. 11. AR-HUD in Mercedes-Benz S-Class (left) and in VW ID.3 (right) for different distances from intersection (approximate distance for Phase 1: 70 m, Phase 2: 50 m, Phase 3: 30 m, Phase 4: 10 m; during Phase 4 of VW ID.3, the chevrons are hardly visible, as they start to fade and are masked by the cars in the background).

4.1.4. Experimental drives, instructions, and procedure

During the whole experiment, the subjects conducted three different drives (Table 3):

The whole experimental procedure lasted 2.5 h and could be divided into five phases:

First, the participant was welcomed and informed about the general course of the experiment. Various documents were presented and signed (e.g. informed consent, privacy statement). The participant entered the vehicle, adjusted seat and inside mirror, and fastened the seat belt. The head-mounted DRT was placed on the subject's head and the micro switch button for the DRT was attached to the left index finger of the participant. The tablet for the secondary task was shown to the participants. The participant performed a short practice trip (identical route as Baseline drive) in order to familiarize with the DRT and the handling of the micro switch button. The phase ended with a Baseline and a Reference drive. Before the Reference drive started, the participant practiced the manual radio tuning task.

Second, the participant experienced one of the two experimental HMI conditions (HUD, AR-HUD; depending on permutation plan) while conducting the Navigation drive. After the drive, the participants gave feedback concerning their workload and filled out a questionnaire measuring user experience and subjective distraction from the driving task caused by the HMI.

Third, the participant repeated the Baseline and the Reference drive.

Fourth, the participant experienced the second experimental HMI condition (HUD, AR-HUD; depending on permutation plan) while conducting the Navigation drive. After the drive, the participants gave feedback concerning their workload and filled out a questionnaire measuring user experience and subjective distraction from the driving task caused by the HMI.

Fifth and after experiencing both HMI variants, the participant again conducted the Baseline and the Reference drive. Moreover, a preference judgement concerning the HMI was asked for and the participant filled out a questionnaire measuring feedback concerning the DRT. Finally, the participant received a compensation and was discharged.

4.1.5. Dependent measures

A number of variables from the areas (1) preference and user experience, (2) workload, and (3) acceptance of DRT was measured (Table 4):

4.1.6. Analyses and statistics

The data were analysed both descriptively and inferential statistically. To assess the effects of the within factor 'Drive' (Baseline vs Reference vs HUD Navigation vs AR-HUD Navigation) and the between factor 'Vehicle' (Mercedes vs Volkswagen), an analysis of variance (mixed ANOVA) with repeated measurements was conducted. If a main effect reaches statistical significance, Scheffé tests were used for post-hoc comparisons (as this is a conservative method of post-hoc testing; Winer, 1962). Mean values (m) and confidence intervals (CI) were calculated for post-hoc comparisons.

Mauchly's test was used for proving sphericity, Box's test for testing homogeneity of covariances, and Leven's test for checking homoscedasticity. Normal distribution of the dependent variables was assumed, since ANOVA is robust to a violation of the normal distribution assumption (Pagano, 2010). Greenhouse-Geisser adjustment was used to correct for violations of sphericity. If there was

Table 3
Description of and instruction for the experimental drives.

	Description	Instruction
Baseline drive	Urban route in an industrial area with right-turn intersections only. Duration: approx. 2 min. HMI: HUD. Subjects drove the course manually and without navigation system and carried out the DRT.	"Adhere to the road traffic regulations and pay attention to my instructions on which direction to turn. Please work on the reaction task. Nevertheless, the driving task should always have the highest priority. Work on the reaction task only if this does not jeopardise your driving safety."
Reference drive	Urban route in an industrial area with right-turn intersections only. Duration: approx. 2 min. HMI: HUD. Subjects drove the course manually and without navigation system, worked on a secondary task (Manual Radio Tuning Task; Krause et al., 2015), and carried out the DRT.	"In addition to the driving and reaction task, you are now to complete another task, the so-called manual radio tuning task. This is to simulate the operation of an "old radio" where you still had to set frequencies yourself. [Detailed description of manual radio tuning task.] Adhere to the road traffic regulations and pay attention to my instructions on which direction to turn. Please work on the reaction task. Nevertheless, the driving task should always have the highest priority, followed by the manual radio tuning task. Work on the reaction task only if this does not jeopardise your driving safety."
Navigation drive	Urban route in an industrial and residential area with different types of intersections (e.g. single lane, right before left; several lanes, traffic controlled; roundabout). Duration: approx. 7 min. HMI: according to experimental condition. Subjects drove the course manually and with navigation system, and carried out the DRT. As participants drove the navigation drive two times, two courses were used with a modified sequence of the crossing scenarios in each case.	"You now drive with the navigation system active. This means that you will receive navigation instructions in the HUD/AR-HUD [depending on HMI condition]. Please use this information and drive according to these instructions. Please work on the reaction task. Nevertheless, the driving task should always have the highest priority. Work on the reaction task only if this does not jeopardise your driving safety."

Table 4
Dependent variables.

Preference and user experience	Preference judgement of the HMI variants: At the end of the experiment, the participants answered the question “Which of the display variants did you prefer to drive with?” [rank]. Standardized questionnaire: The participants filled out a standardized questionnaire on user experience after each condition (UEQ-S; Schrepp et al., 2017). The questionnaire includes two subscales: pragmatic (task related aspects, e.g. efficiency, perspicuity) and hedonic quality (non-task related aspects, e.g. stimulation, novelty). Obstruction and distraction by display: After each condition, the respondents rejected or agreed on two statements concerning obstruction and distraction by the display (“The display obstructed important areas of the road.”, “The display distracted me from my driving task.”) [agreement to statement on five-point rating scale].
Workload	Standardized questionnaire: The participants filled out a standardized questionnaire measuring workload after each drive (DALI; Pausie, 2008). Detection response task: General degree of visual-cognitive distraction measured by the parameters miss rate [% visual targets not detected] and reaction time for detected visual targets [ms].
Acceptance of DRT	Feedback concerning DRT: At the end of the experimental procedure, participants answered several questions regarding their experience with performing the DRT (“In general, the stimulus was badly perceptible.”, “Working on the reaction task was unpleasant.”, “I would not participate again in a similar experiment.”) [agreement to statement on five-point rating scale].

no homogeneity of the error variances, post-hoc tests were calculated to confirm the results of the ANOVA.

The measures for the Baseline drive and the Reference drive were averaged over the three repetitions (in the beginning, middle, and end of the experimental procedure). An alpha level of 0.05 was used to determine statistical significance.

4.2. Results

4.2.1. Preference and user experience

After experiencing both HMI conditions (HUD and AR-HUD) in the Volkswagen ID.3 or Mercedes S-Class, the subjects were asked which HMI condition they would prefer. For both vehicles, most participants (VW: $n = 9$, i.e. 75 %; MB: $n = 9$, i.e. 75 %) preferred the AR-HUD (Fig. 12 left).

The user experience was significantly better for the AR-HUD compared to the HUD ($F(1, 22) = 4.99$, $p < .036$). This significant main effect of the HMI condition was solely due to the better hedonic quality of the AR-HUD ($F(1, 22) = 21.85$, $p < .001$; Fig. 12 right). Because there was no homogeneity of the error variances, as assessed by Levene's test ($p < .05$), post-hoc tests were calculated to confirm the results ($m_{AR-HUD} = 1.93$, 95 %-CI_{AR-HUD}[1.52, 2.36]; $m_{HUD} = 0.63$, 95 %-CI_{HUD}[0.16, 1.11]). The pragmatic quality did not differ between HUD and AR-HUD ($F(1, 22) = 0.12$, $p = .728$; Fig. 12 middle).

Both vehicles differed only in terms of hedonic quality with better ratings for the Mercedes S-Class ($F(1, 22) = 6.09$, $p = .022$). Again, post-hoc tests were calculated to confirm the results ($m_{VW} = 0.88$, 95 %-CI_{VW}[0.39, 1.36]; $m_{Mercedes} = 1.70$, 95 %-CI_{Mercedes}[1.21, 2.19]). There were no significant interactions for HMI $\times$ Vehicle.

After experiencing each HMI condition, participants reported to what extent they agree with the statements that the display obstructed important areas of the road or distracted from the driving task. Statistical analyses showed a significant main effect of HMI on obstruction ($F(1, 21) = 5.32$, $p = .031$) and distraction ($F(1, 21) = 5.34$, $p = .031$) and no significant effect of vehicle. The AR-HUD distracted a bit more than the HUD. The answers show (Fig. 13) that the HUD did neither obstruct important areas of the road nor distract from driving. Concerning the AR-HUD, one participant driving the Mercedes S-Class reported obstruction and distraction and two participants driving the Volkswagen ID.3.

4.2.2. Workload

Results showed a significant main effect of ‘Drive’ ($F(3, 66) = 21.85$, $p < .001$) on subjective workload. Post-hoc testing indicated that the Reference drive was more demanding than the Baseline drive and the workload of the Navigation drives was in between

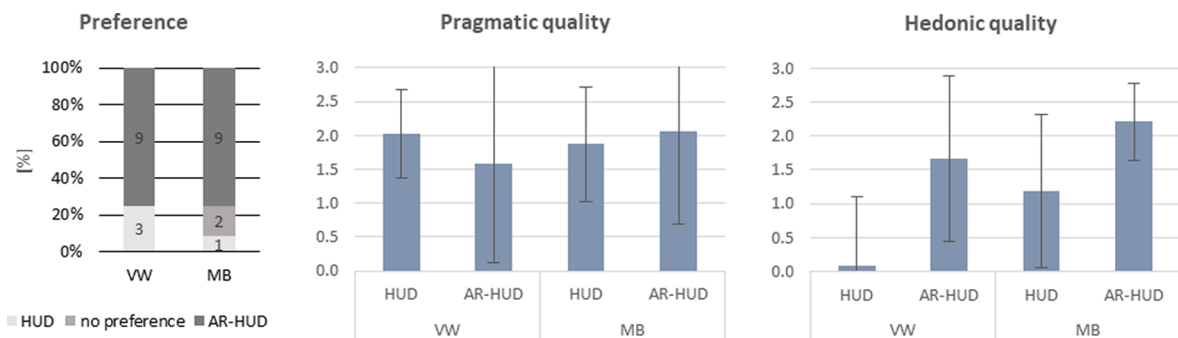


Fig. 12. Preference of HMI conditions (left), rating of pragmatic quality of user experience (middle) and hedonic quality of user experience (right) for HMI conditions (mean value $\pm$ standard deviation).

($m_{\text{Baseline}} = 24.72$, 95 %-CI_{Baseline}[19.90, 29.55]; $m_{\text{Reference}} = 59.72$, 95 %-CI_{Reference}[50.77, 68.67]; $m_{\text{NavigationAR-HUD}} = 42.67$, 95 %-CI_{NavigationAR-HUD}[32.93, 52.40]; $m_{\text{NavigationHUD}} = 39.00$, 95 %-CI_{NavigationHUD}[30.66, 47.34]; Fig. 14 top). The HMI condition (HUD vs AR-HUD) did not lead to any change in subjective workload.

A very similar result is found for the DRT misses (Fig. 14 bottom left). Again, a significant main effect ($F(3, 63) = 31.51$, $p < .001$) and subsequent post-hoc tests (also necessary because of missing homogeneity of the error variances) revealed the highest workload for the Reference drive, followed by the Navigation drives. The lowest workload was found for the Baseline drive. There was no significant difference between the HMI conditions of the Navigation drive ($m_{\text{Baseline}} = 0.11$, 95 %-CI_{Baseline}[0.06, 0.16]; $m_{\text{Reference}} = 0.40$, 95 %-CI_{Reference}[0.31, 0.48]; $m_{\text{NavigationAR-HUD}} = 0.23$, 95 %-CI_{NavigationAR-HUD}[0.14, 0.31]; $m_{\text{NavigationHUD}} = 0.21$, 95 %-CI_{NavigationHUD}[0.16, 0.27]).

Reaction times showed also the significant main effect ($F(3, 63) = 42.53$, $p < .001$) with post-hoc testing indicating longer reaction times and higher workload for the Reference drive compared to the Baseline drive or the Navigation drives (Fig. 14 bottom right). The Navigation drives with HUD showed comparable reaction times as in the Baseline drive. The Navigation drives with AR-HUD were characterised by longer reaction times than during the Baseline drive. During all Navigation drives, participants responded faster to the DRT stimuli than during the Reference drive. Again, there was no significant difference between the HMI conditions of the Navigation drive ($m_{\text{Baseline}} = 618$, 95 %-CI_{Baseline}[573, 663]; $m_{\text{Reference}} = 847$, 95 %-CI_{Reference}[794, 900]; $m_{\text{NavigationAR-HUD}} = 719$, 95 %-CI_{NavigationAR-HUD}[655, 783]; $m_{\text{NavigationHUD}} = 676$, 95 %-CI_{NavigationHUD}[623, 730]).

4.2.3. Learning effect

The DRT workload parameters, i.e. DRT miss rate (Fig. 15 top left) and DRT reaction time (Fig. 15 top right), do not indicate a clear learning effect, because there was no significant effect of repetition (i.e. if the participant conducted the Baseline drive or Reference drive for the first, second, or third time; DRT miss rate: $F(2, 36) = 0.74$, $p = .483$; DRT reaction time: $F(2, 36) = 2.56$, $p = .091$).

In contrast to the DRT workload parameters, the subjective workload measured by the DALI questionnaire (Fig. 15 bottom left) showed a significant learning effect with lower subjective workload when participants repeated the drive (Greenhouse-Geisser adjustment was used to correct for violations of sphericity: $F(1.55, 34.11) = 35.33$, $p < .001$).

Furthermore, repeated Reference drives were characterised by an increasing number of secondary tasks completed by the subjects (Fig. 15 bottom right).

4.2.4. Acceptance of DRT

At the end of the experiment, some participants ($n = 7$; 29 %) reported that the DRT was not easy to perceive (Fig. 16). The perceptibility of the stimulus did not depend on the weather conditions (sunny, cloudy, rain). Five subjects (21 %) rather agreed to the statement that the DRT was unpleasant. Nevertheless, all test persons would participate again in a similar experimental study.

5. Discussion and conclusions

Many studies evaluated the effects of automotive HUDs compared to HDD, mostly resulting in beneficial effects for HUDs (Horrey et al., 2003; Kiefer, 1991, 1998; Kim et al., 2013; Lorenz et al., 2014; Sprenger, 1993). Although several studies also resulted in positive effects of AR-HUDs (Bark et al., 2014; Bauerfeind et al., 2019; Bauerfeind et al., 2021; Frémont et al., 2020; Kim et al., 2016; Medenica et al., 2011; Rusch et al., 2013; Schall et al., 2013; Wang et al., 2020), it remains unclear how AR-elements have to be designed to prevent potential negative side-effects like distraction and additional workload.

This paper evaluated a methodology based on the DRT (ISO, 2016) for measuring the distraction and workload effect of three different display technologies: HDD, HUD, and AR-HUD. The DRT methodology is based on a recognition-response task and results in two measures that quantify the level of cognitive and visual workload: reaction time and miss rate. The main goal was to derive to what extent the DRT method is suitable to measure visual-cognitive workload in road traffic and what to consider when using it during experimental studies in the driving simulator and in real traffic environments. Furthermore, guidelines for the practical use of the method should be derived.

Two experimental studies in the driving simulation and in real traffic evaluated the workload effects of AR-HUDs displaying navigation instructions while driving through an urban traffic environment with different crossing scenarios. The results of the experimental studies show that drivers prefer the AR-HUD (RQ Ia). This is mainly due to the hedonic quality of the AR-HUDs and is

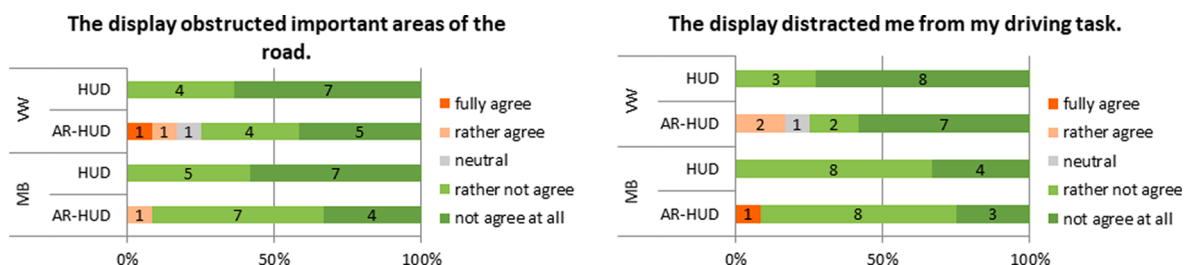


Fig. 13. Feedback of drivers concerning obstruction (left) and distraction (right) by display.

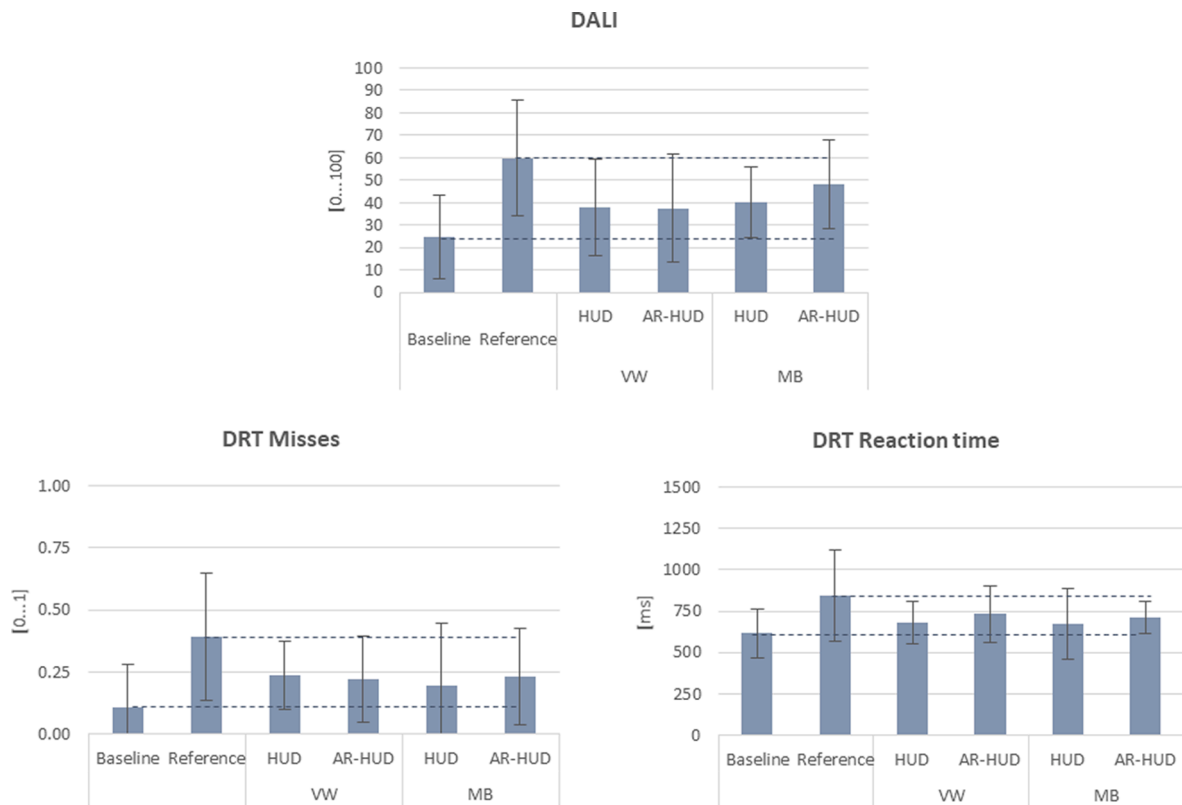


Fig. 14. Subjective workload (top), DRT miss rate (bottom left), and DRT reaction time (bottom right) for Baseline drive, Reference drive and HMI conditions in Navigation drive (mean value $\pm$ standard deviation).

consistent with the results of other studies (Bark et al., 2014; Medenica et al., 2011). Nevertheless, drivers reported that AR-elements could obstruct important areas of the road and distract from the driving task. An effect that could be at least partially comparable to the phenomenon of change blindness (O'Regan et al., 1999).

The subjective and visual-cognitive workload measured by the DRT did not differ between the HMI conditions (RQ Ib). This result is similar for both experimental studies, i.e. Study 1 in the driving simulator as well as Study 2 in real traffic.

Nevertheless, an explorative analysis for different types of intersections in Study 1 resulted in more DRT misses for the AR-HUD condition (RQ Ic). The reason why participants had poorer results in the DRT task when driving with AR-HUD remains unclear and should be subject for further research. Possible explanations are that the AR-HUD elements (a) attract attention directly (e.g. by their salience; O'Regan et al., 1999) and/or (b) obstruct important areas of the road which in turn demands drivers to direct more attention to the driving environment.

A higher number of DRT misses while in approaching and driving through the intersections in the AR-HUD condition indicates a higher workload. This is also reflected by a later start of blinking in front of the intersection (RQ Id). However, this occurs in favor of an increase in the navigation task performance as no driver made a wrong turning decision in the AR-HUD condition. This facilitating effect on drivers' navigation task was also measured by Bauerfeind et al. (2019).

The DRT clearly shows a difference in visual-cognitive workload between Baseline drive, Navigation drive, and Reference drive. An approach that was similar to the one applied by Strayer et al. (2017). Furthermore, the results of Study 1 in the driving simulator revealed that the DRT is able to assess aspects of visual workload of HUD displays which are independent from the eyes-off-road time (RQ IIa): Although drivers do not have to divert their attention away from the road (as in the HDD condition) their subjectively perceived workload level as well as the ability to detect and react to the DRT presented in the driving scene is equally impaired by the different HMI conditions – if not even more impaired by the AR-HUD. Concerning the application of the DRT, the results indicate that the global DRT performance across the complete test course seems to be less sensitive for finding differences between experimental conditions than the analysis of only those sections that are directly linked to the intersections.

The results of Study 2 in the real vehicles revealed an obvious learning effect in the subjective workload measure but hardly in objective DRT measures (RQ IIb). The reason of this is twofold: First, during the Baseline drives a statistical floor effect hindered the measurement of improved performance in participants' reaction to the DRT. Reaction times of approx. 0.65 s during the first Baseline drive were already very fast and participants were not able to react even faster when repeating the drive in the course of the experiment (especially as they drive in a real traffic environment). Second, with repeating Reference drives the subjects carried out an increasing number of secondary tasks. That means, that participants used their experience by improving their performance in the manual radio

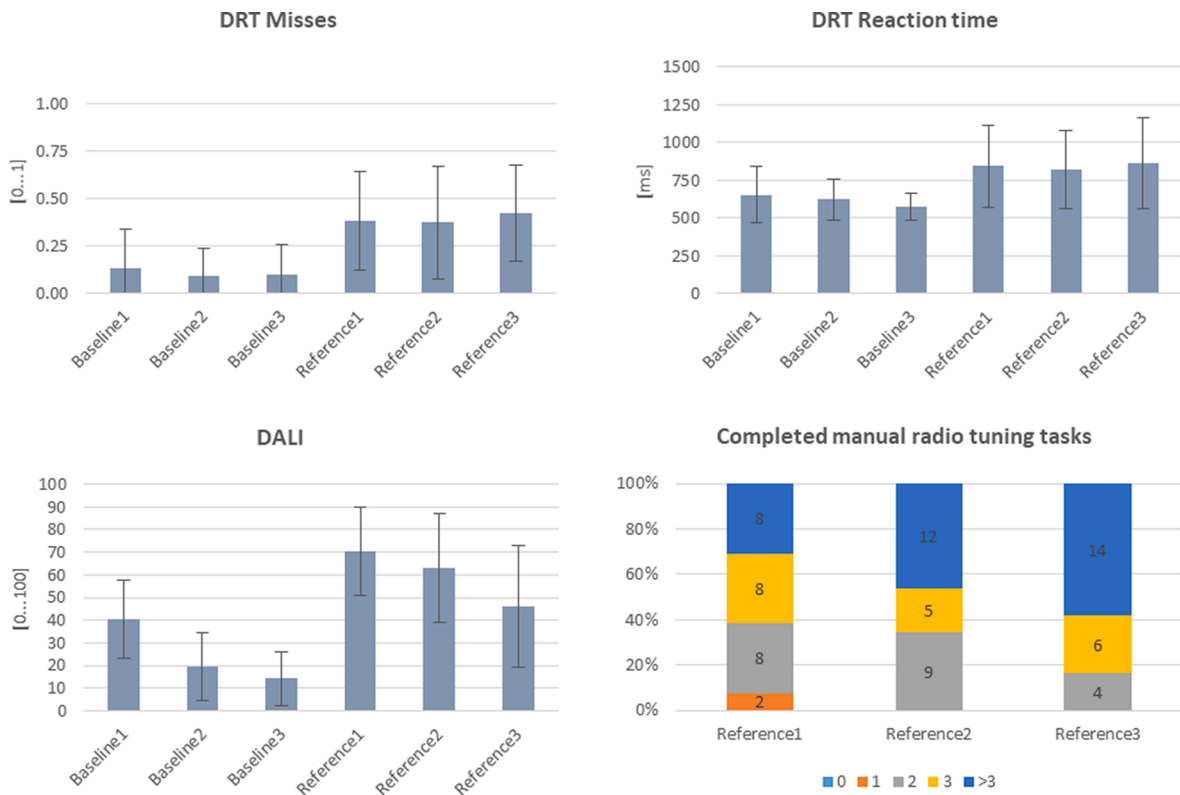


Fig. 15. DRT miss rate (top left), DRT reaction time (top right), subjective workload (bottom left), and number of completed manual radio tuning tasks (bottom right) for Baseline and Reference drives (mean value $\pm$ standard deviation).

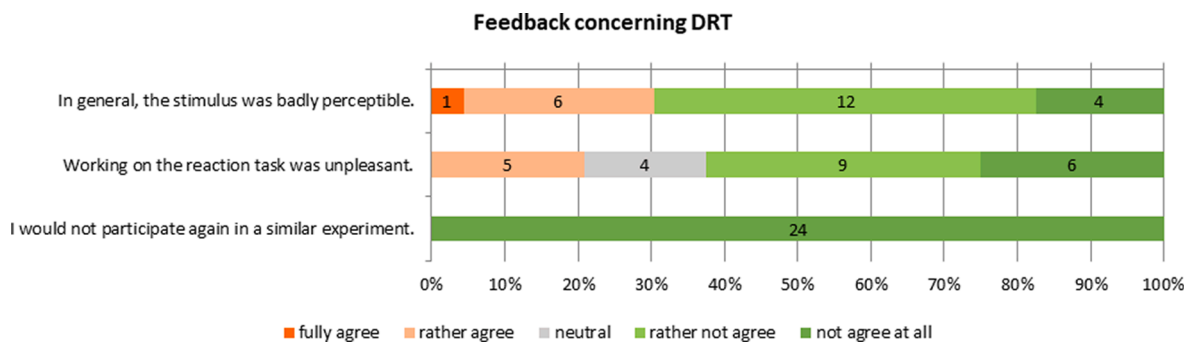


Fig. 16. Feedback of drivers concerning DRT method.

tuning task (and not in the DRT). This was in line with the given instruction for prioritising the tasks.

Participants accepted the DRT as a method for workload assessment (RQ IIC). Therefore, the two DRT measures miss rate and reaction time are a valid and objective measure of drivers' visual-cognitive workload while using AR-HUD based HMIs (Stojmenova, & Sodnik, 2018).

Table 5 gives guidelines how to use the DRT method for measuring visual-cognitive workload while driving (based on the ISO 17,488 as well as the experience gained by conducting the subject studies in the driving simulator and in real traffic).

The HMI and the displays of both tested vehicles differed in many aspects with the Mercedes showing a more premium look and feel compared to the Volkswagen. Therefore, subjects rated the Mercedes as having a higher hedonic quality. The AR-HUD of the Volkswagen is a dot-matrix compared the more brilliant AR-HUD of the Mercedes that also contains much more information. Nevertheless, the subjective workload as well as the DRT measures did not reveal significant differences between both vehicles. Drivers seem to be able to focus on the relevant information for driving without being distracted by additional elements in the HUD.

The studies in the driving simulator and in real traffic showed some differences. For example, the DRT reaction time during the Baseline drive is not only greater in the real world compared to the simulator but also more dispersed. Nevertheless, more detailed

comparisons between real world and simulation would have exceeded the scope of the current paper. Further issues which are relevant for practical application remain open and should be addressed in future research:

- (1) To what extent do the results depend on the specific use case applied during the study, i.e., the driving task that the subjects had to perform under different HMI conditions (e.g. navigating through an urban environment vs longitudinal or lateral driving performance)?
- (2) How sensitive is the DRT to evaluate different design aspects of AR-HUDs? This issue could be analysed by varying design aspects of the AR-HUD (e.g. the size or dynamics of the information displayed, the timing of the display) and measuring effects on DRT parameters.
- (3) How suitable is the usage of a tactile DRT to measure distraction and workload caused by AR-HUDs? A potential advantage of the tactile DRT could be the focus on attentional effects of cognitive workload, i.e. without the influence of visual workload.
- (4) Is the DRT able to distinguish between visual and cognitive workload? One approach to answer this could be the targeted variation of visual workload and/or cognitive workload while driving.
- (5) Which secondary tasks, which are carried out while driving during the reference drive, are appropriate (as an alternative to the Manual Radio Tuning Task used during the experimental studies reported in this paper)?
- (6) Which data analysis strategy is most suitable to show differences between AR-HUD concepts? This includes the influence of the analysed time window (especially its size, i.e. the length of the section used for analysis) on the results and the sensitivity of finding effects?

Overall, the DRT is a promising approach to measure and quantify workload effects of traditional HUDs compared to displays using AR-elements. As these AR-HUDs will probably become more frequent, the DRT is a valuable method for measuring potential negative side-effects of AR-HUDs on workload and distraction in an objective way and continuously while driving. By this, human factors experts can give useful instructions and suggestions to developers on how to design AR-HUDs (e.g. in terms of number of elements, colour, size, dynamics and movement, display duration).

CRedit authorship contribution statement

Christian Maag: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Nadja Schömig:** Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Frederik Naujoks:** Conceptualization, Methodology, Validation, Writing – review & editing. **Ines Karl:** Conceptualization, Methodology, Validation, Writing – review & editing. **Andreas Keinath:** Supervision. **Alexandra Neukum:** Resources, Supervision, Project administration, Funding acquisition.

Table 5

Guidelines for applying DRT for measuring visual-cognitive workload while driving.

Applicability of the DRT for different research questions	For studies of driver distraction that look at aspects of visual-cognitive distraction which is not measurable via gaze analyses (e.g., in studies of AR-HUD) DRT is the method of choice. Depending on the research focus, an appropriate DRT variant should be selected: Head-mounted DRT: Assessment of visual-cognitive workload (1) when gaze direction requirements are not of interest and (2) for use cases with many head turns (e.g., turning operations in real traffic). Tactile DRT: Assessment of cognitive workload (i.e., without visual components). Remote DRT: Assessment of visual-cognitive workload and relevance of gaze direction (e.g., when evaluating different display positions).
Methodological aspects	The DRT is suitable for studies in the driving simulator as well as in real traffic. Caution: Driving activities with a manual component (e.g., turning) can impair the DRT performance. Since carrying out the DRT is strenuous for study participants, the route sections should not be too long (i.e., no longer than 15–20 min). A Baseline and a Reference drive are necessary to allow relative assessment of DRT performance. One possible reference task is the Manual Radio Tuning Task (Krause et al., 2015) as a worst case scenario. When instructing the subject, the different subtasks have to be prioritised: Driving has highest priority, followed by the DRT and secondary task (if any). The experimenter should emphasize that the experiment is not testing participants' abilities but the extent to which different driving or secondary tasks affect performance. The sample should include at least 24 subjects, as required by NHTSA (2013), and divided into 4 age groups (18–24, 25–39, 40–55, >55 years). A preliminary study with at least 3 naïve subjects should be conducted. Because of expected learning and fatigue effects, an appropriate permutation schedule should be established. The experimental procedure should include extensive training (as specified in ISO 17488) to minimize learning effects in the study. Baseline and Reference drives could be repeated to quantify further learning effects.
Analysis	After completion of the data collection, the various workload parameters are evaluated. This includes e.g. subjective workload (using questionnaire), visual workload (i.e. eye tracking), and visual-cognitive workload using DRT (miss rate, reaction time for valid hits). The route influences the evaluation strategy: If the experimental treatment influences the driver's workload during most trip sections, the entire drive can be analysed. If the experimental treatment presumably influences the workload only in selected scenarios, the analysis should focus on these. The meaning of the workload measured during the treatment condition is relative, e.g. it should be compared to Baseline and Reference runs or other treatment conditions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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